ANUJ POKHAREL

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PROFESSIONAL SUMMARY

Software Engineer with 6 years building production systems across SaaS, fintech, and e-commerce. Specialized in backend architecture (Django/FastAPI/Node J's), cloud infrastructure (AWS), and ML pipelines (TensorFlow/PyTorch).Proven track record collaborating with cross-functional teams and international clients. Seeking senior engineering roles at innovative tech companies.

TECHNICAL SKILLS

Languages: Python, JavaScript/TypeScript, SQL
Backend: Django, Django REST Framework, FastAPI,Node J's, REST
Frontend: React.js
Databases: PostgreSQL, MongoDB, MySQL, Redis
ORM: Django ORM, SQLAlchemy
AI Architecture: RAG (Retrieval-Augmented Generation), Multi-Agent Systems
ML/AI Libraries & Tools: TensorFlow, PyTorch, scikit-learn, Hugging Face, LangChain, OpenCV, NLTK, spaCy
Voice AI: Deepgram,Whisper, ElevenLabs, Cartesia
Cloud/DevOps: AWS (EC2, S3, RDS, Lambda), Hetzner, Docker, Kubernetes, Nginx, CI/CD, GitHub Actions
Tools/Platforms: Git, Celery, Redis, Stripe API, Twilio, Postman, JIRA, Expo, GitLab, GitHub
API Tech & Documentation: GraphQL, Swagger, Postman
Testing: pytest, Jest, Selenium, unittest, integration testing
Methodologies: Agile/Scrum, REST APIs, Microservices, TDD, System Design

PROFESSIONAL EXPERIENCE

Senior Agentic AI Engineer, CTO & Co-Founder Aug 2025 – July 2026
Codex Neural Remote

- Co-founded **Codex Neural** and lead technology strategy and engineering as CTO, overseeing the design and delivery of agentic AI products for the life sciences industry
- Architected a **multi-agentic AI system** for lfescienceaiagents.com, orchestrating specialized autonomous agents to support research, discovery, and operational workflows in the life sciences domain
- Directed end-to-end technical roadmap, from system architecture and agent orchestration frameworks to deployment infrastructure, as the company's founding technical leader

Senior AI Engineer (Part-Time) Jan 2026 – Present
Mavens Solutions Kathmandu, Nepal

- Architecting a real-time conversational voice AI model for a fitness application, integrating **Deepgram** for speech-to-text transcription and **ElevenLabs** and **Cartesia** for natural, low-latency text-to-speech synthesis
- Designed a **RAG (Retrieval-Augmented Generation)** pipeline using **Hugging Face** models to ground voice responses in fitness domain knowledge, enabling context-aware coaching and Q&A
- Evaluating and benchmarking multiple TTS providers (ElevenLabs vs. Cartesia) to optimize for voice naturalness, latency, and cost-efficiency in a production voice pipeline

Back-End Developer (Contract) Feb 2025 – Aug 2025
Neatly – On-demand Cleaning Marketplace SaaS Remote, Georgia USA

- Architected and deployed backend microservices using **Django + PostgreSQL** serving 5K+ active users; designed normalized database schemas reducing query time by 40%
- Implemented asynchronous task processing with **Celery + Redis** for image optimization, email notifications, and scheduled jobs; reduced API response time from 800ms to 200ms
- Integrated **Stripe payment gateway** and **Twilio OTP authentication**, processing $50K+ monthly transactions with zero payment failures
- Built cross-platform mobile app using **React Native** with camera integration, geolocation services, and push notifications; achieved 4.5+ app store rating
- Optimized database queries and added composite indexes, improving dashboard load time by 60%

Full Stack Developer (Contract) Aug 2024 – Nov 2024
Codeilo / Codelio Solution Remote

- Developed multi-tenant SaaS ERP system using **Next.js** and **Django, DRF, PostgreSQL**; implemented role-based access control (RBAC) supporting 20+ permission levels
- Designed and implemented **RESTful APIs** with JWT authentication, rate limiting, and comprehensive error handling; maintained 99.5% uptime
- Built subscription billing system with recurring payments, usage tracking, and automated invoicing
- Containerized applications using **Docker** and set up **CI/CD pipelines** with GitHub Actions, reducing deployment time from 2 hours to 15 minutes
- Key projects delivered: OurCEO, OurBriefcase

Software Developer & ML Consultant (Freelance) Jun 2021 – Present
Self-Employed Remote

- Delivered 15+ client projects including e-commerce platforms, job portals, and document management systems using **Django/FastAPI/React/React Native** tech stack
- Built NLP models for spam detection (95% accuracy) and Nepali-English translation using **transformer architectures** and Hugging Face
- Developed computer vision solution for counterfeit currency detection using **CNNs**; achieved 92% precision on test dataset
- Implemented **RAG-based chatbot** using LangChain and OpenAI API for customer support automation, reducing response time by 70%

KEY PROJECTS

Counterfeit Currency Detection System | ***TensorFlow, OpenCV, Python, FastAPI*** 2024

- Led team of 3 to develop CNN-based classification system detecting fake currency notes with 92% accuracy
- Created custom dataset of 5000+ images with data augmentation; implemented transfer learning with VGG16
- Deployed REST API serving 100+ requests/day; implemented model monitoring and versioning

River Water-Level Forecasting System | ***Python, SARIMA, VAR, PostgreSQL*** 2024

- Built production-grade forecasting pipeline with automated ETL, feature engineering, and model retraining
- Achieved 85% accuracy on 7-day forecasts; implemented real-time alerting system for flood warnings

HAMANIO E-Commerce Platform | ***Django, React, Redis, Celery, LangChain*** 2025

- Developed full-stack marketplace with product catalog, shopping cart, payment integration, and order management
- Implemented background job processing with Celery; integrated RAG-powered chatbot for customer support

EDUCATION

Bachelor of Computer Engineering 2019 – 2023
Kathford Engineering College, Tribhuvan University Lalitpur, Nepal

- **Relevant Coursework**: Data Structures & Algorithms, Machine Learning, Database Systems, Computer Networks, Operating Systems, Software Engineering, Cloud Computing

CERTIFICATIONS & AWARDS

Front-End Development with React – Coursera (Meta)
AWS Certified Cloud Practitioner – In Progress
Academic Excellence Award – Kathford Engineering College, 2022

ADDITIONAL INFORMATION

Languages: English (Professional), Nepali (Native), Hindi (Conversational)
Interests: Open-source contribution, Competitive Programming, AI Research, Tech Blogging